PV102 Day 7 Student Packet

Installation Planning Mechanical Layout and JHA

PV102 | CTS Day 23 | Student Print Packet
Plan the install before lifting parts. | Wednesday, August 12, 2026

Today's rule

Plan the install before lifting parts.

Learning targets

- Convert a design into a sequenced work plan.
- Identify mechanical, electrical, fall, weather, access, and tool hazards.
- Build a material/tool list that matches the scope.
- Use checklists to prevent skipped steps and hidden coupling.

Morning quiz

1. A JHA is used to:

- A. Identify hazards and controls before work
- B. Replace all training
- C. Make electricity harmless
- D. Skip PPE

2. A work sequence should include:

- A. Setup, controls, task steps, checks, cleanup
- B. Only the fastest path
- C. Only tool names
- D. No safety boundary

3. The best time to plan material staging is:

- A. Before work starts
- B. After parts are missing
- C. During an emergency
- D. Never

4. A wire route should consider:

- A. Protection, distance, access, and serviceability
- B. Wall color only
- C. The shortest random path
- D. Nobody knows

5. A quality hold point means:

- A. Stop and verify before covering or continuing
- B. Stop learning
- C. Hide errors
- D. Energize faster

6. Name one mechanical hazard in PV installation planning.

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7. Name one electrical hazard that remains even before final commissioning.

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8. Why should tools and materials be staged instead of scattered?

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9. What information from Day 2 changes the install plan?

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10. Write the Day 7 rule in your own words.

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_____

Reference: decision order

Step	Action	Proof question
1	Define the job	What must the system accomplish?
2	Collect evidence	What is measured, observed, or still unknown?
3	Choose boundary	What must be protected, isolated, sized, or verified?
4	Defend the choice	What evidence supports the decision?
5	Handoff	What should the next person know?

Install Plan and Job Hazard Analysis

Teams create a one-day install plan for a training system and defend the safety controls before work begins.

Scenario options

- Tabletop PV training kit build: no energized connections until inspection.
- Mock roof rack layout: rails, clamps, wire management, no live module connection.
- Bus conversion equipment board: battery/inverter/controller placement and service clearances.

Team procedure

- Choose a scenario and define the scope in one sentence.
- Create a sequenced work plan: setup, layout, mounting/staging, wiring route, checks, cleanup.
- Complete a JHA with hazards, controls, PPE, and stop-work triggers.
- Build a material/tool list and mark missing items.
- Add at least three quality hold points.
- Handoff the plan as if a substitute crew had to run it safely.

Team worksheet

Prompt	Team response
System/job statement	
Evidence used	
Decision made	
Safety boundary	
Top unknowns	
Next action	

Exit ticket

What is one step your team added that prevents a mistake before it becomes expensive or unsafe?

_____

_____